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Research Lab: NEWS Lab
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Updated 2026-11-03 (Self)

Dr. Alexander C. Nwala

Research interests

Social Media/Computational Social Science, Web/Data Science
Web Archiving, (Local)News, NLP

Education

2020 – 2022 **Postdoctoral**, *Observatory on Social Media, Indiana University, Bloomington, Indiana*, USA
Advisor: Filippo Menczer

2014 – 2020 **Ph.D. Computer Science**, *Old Dominion University (ODU), Norfolk, Virginia*, USA
Thesis: Bootstrapping Web Archive Collections From Micro-Collections In Social Media,
Summa Cum Laude, Advisor: Michael Nelson

2012 – 2014 **M.S. Computer Science**, *Old Dominion University, Norfolk, Virginia*, USA
Thesis: Generating Combinatorial Objects - A New Perspective,
Summa Cum Laude, Advisor: Stephan Olariu

2008 – 2011 **B.S. Computer Science (Minor: Aviation Science)**, *Elizabeth City State University, Elizabeth City, North Carolina*, USA
Summa Cum Laude

Academic Appointments

08/2022 – Present **Assistant Professor, William & Mary, Williamsburg Virginia**

08/2020 – 07/2022 **Postdoctoral Researcher, Observatory on Social Media, Indiana University**

Grants ($980K total, Awarded: 6, PI: 4, External: 3)

2026 (Awarded $80K, W&M: $40K): COVA CCI: Securing Agentic AI Systems using Verified Behavioral Templates
2026 (Awarded $750K): IMLS: CiteCast: Linking US local & national television news to cited web resources
2023 – 2026 (Awarded $175K): NSF CRII: III: Measuring the alignment between the interests of local communities, local news, and the national news media
2022 – 2024 (Awarded $10K): W&M Faculty Research Grants
2022 – 2023 (Awarded Co-PI,$5K): W&M Arts and Science Dean's Faculty Support

Teaching

William & Mary

Key: U **- Undergraduate,** G **- Graduate,** N **- New,** D **- Designed,** R **- Restructured**

Spring/Fall 2026 **Reasoning with GenAI - DATA 101**^{UR}

Fall 2024 **Ethics in Data Science - DATA 202-01**^{UR}

Spring 2024 **Network Analysis - DATA 641-01**^{GND}**/445-01**^{UND}

Spring 2023 **Network Science - DATA 340-02**^{UND}

Fall 2022/2023/2024 **Web Science - DATA 691-03**^{GN}**/440-{02,03,07}**^{UN}

Old Dominion University

Spring 2018/2019 **Web Science - CS 432**^{U}**/532**^{G}

July - November 2018 **Introduction to Programming with C++ - CS 150**
Volunteer C++ Instructor for Middle/High School Students (ages 11 - 16)

Fall 2014, **Programming and Problem Solving in C++ - CS 333/CS 250**
Spring/Fall 2015 Teaching Assistant/Lab Instructor

Advising at William & Mary

Undergraduate

- **Major Advisor:** Fall 2026: 15 students, Spring 2026: 11 students, Fall 2025: 12 students, Spring 2025: 17 students, Fall 2024: 17 students, Spring 2024: 3 students, Fall 2023: 2 students, and Spring 2023: 2 students.
- **Primary advisor for Honors' thesis:** Fall 2022 – Spring 2023: Metzger Zoe (Identifying Social Media Users that are Susceptible to Phishing Attacks)
- **Honors' committee member:** Fall 2023: Seggelke Kayla and Larimer Sarah. Spring 2024: Crowley Kaitlyn.
- **Primary advisor for Monroe scholar:** 2024: Kim Hannah (How do makeup trends propagate and evolve across TikTok)
- **Undergraduate research group members:** 2022: Nelson Ethan and Huang Sofia 2023: Swack Jeremy, Nelson Ethan, and McDonalds Ian. 2024: McDonalds Ian and Emmanuel-King Greatness.

Graduate

- **Current PhD Students:** Ariyarathne Gangani and Ariyarathne Isuru.

Software/Services

- agwagram: Social media behavior explorer
- Coordiscope: Inauthentic coordination explorer
- BLOC: Social media user behavior model
- Sumgram: Text collection summarization with most frequent conjoined ngrams
- StoryGraph: Real-time news Story Link Detection (Twitter Bot, Github)
- Local Memory Project: US/non-US local news media sources

Google Scholar

Citations: 271, h-index: 9 (at 2026-11-03): https://scholar.google.com/citations?user=LqrUey4AAAAJ&hl=en
Publication Key: *U* - William & Mary Undergraduate author, *G* - William & Mary Graduate author

Publications - Book Chapters/Journals

[1] Kai-Cheng Yang, Onur Varol, Alexander C. Nwala, Mohsen Sayyadiharikandeh, Emilio Ferrara, Alessandro Flammini, and Filippo Menczer. Social bots: Detection and challenges. In Taha Yasseri, editor, *Handbook of Computational Social Science*, chapter 33, pages 473–491. Edward Elgar Publishing Limited, 2025.

[2] James Giroux, Gangani Ariyarathne, Alexander C Nwala, and Cristiano Fanelli. Unmasking social bots: how confident are we? *EPJ Data Science*, 14(1):18, 2025.

[3] Nwala, Alexander C, Alessandro Flammini, and Filippo Menczer. A language framework for modeling social media account behavior. *EPJ Data Science*, 12(1):33, 2023.

[4] Kai-Cheng Yang, Onur Varol, Nwala, Alexander C, Mohsen Sayyadiharikandeh, Emilio Ferrara, Alessandro Flammini, and Filippo Menczer. Social bots: detection and challenges. Preprint 2312.17423, arXiv, 2023. (This is a draft of the chapter. The final version will be available in the Handbook of Computational Social Science edited by Taha Yasseri, forthcoming 2024, Edward Elgar Publishing Ltd. The material cannot be used for any other purpose without further permission of the publisher and is for private use only).

[5] Eoin C Whelan, Nwala, Alexander C, Christopher Osgood, and Stephan Olariu. Selective mutation accumulation: a computational model of the paternal age effect. *Bioinformatics*, 32(24):3790–3797, 2016.

[6] Wu He, Ashish Kshirsagar, Nwala, Alexander, and Yaohang Li. Teaching information security with workflow technology–a case study approach. *Journal of Information Systems Education*, 25(3):201–210, 2014.

Publications - Refereed Conference Proceedings

[1] Isuru Ariyarathne^{G}, Gangani Ariyarathne^{G}, Alessandro Flammini, Filippo Menczer, and Nwala, Alexander C. Behavior Change as a Signal for Identifying Social Media Manipulation. In *Proceedings of the 18th ACM Web Science Conference (WebSci '26)*. ACM, 2026.

[2] Gangani Ariyarathne[G], Isuru Ariyarathne[G], Greatness Emmanuel-King, Kate Lawal, and Nwala, Alexander C. Identifying the Geographic Foci of US Local News. In *Proceedings of the 18th ACM Web Science Conference (WebSci '26)*. ACM, 2026.

[3] Ozgur Can Seckin, Manita Pote, Alexander Nwala, Lake Yin, Luca Luceri, Alessandro Flammini, and Filippo Menczer. Labeled datasets for research on information operations. In *Proc. Intl. AAAI Conf. on Web and Social Media (ICWSM)*, volume 19, pages 2567–2574, 2025.

[4] Gangani Ariyarathne[G] and Nwala, Alexander C. 3DLNews: A three-decade dataset of us local news articles. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24)*, pages 1–5, New York, NY, USA, 2024. ACM.

[5] Christopher Torres-Lugo, Manita Pote, Nwala, Alexander, and Filippo Menczer. Manipulating twitter through deletions. In *Proceedings of AAAI Conference on Web and Social Media (ICWSM)*, pages 1029–1039, 2022.

[6] Nwala, Alexander C, Michele C Weigle, and Michael L Nelson. Garbage, Glitter, or Gold: Assigning Multi-dimensional Quality Scores to Social Media Seeds for Web Archive Collections. In *Proceedings of ACM/IEEE Joint Conference on Digital Libraries (JCDL 2021)*, pages 80–89, 2021.

[7] Dhruv Patel, Nwala, Alexander C, Michele C Weigle, and Michael L Nelson. What Did It Look Like: A service for creating website timelapses using the Memento framework. In *Proceedings of ACM/IEEE Joint Conference on Digital Libraries (JCDL 2021)*, pages 340–341, 2021.

[8] Nwala, Alexander C, Michele C Weigle, and Michael L Nelson. 365 Dots in 2019: Quantifying Attention of News Sources. In *Proceedings of Computation + Journalism Symposium 2020 (CJ 2020)*, 2020.

[9] Yasith Jayawardana, Nwala, Alexander C, Gavindya Jayawardena, Jian Wu, Sampath Jayarathna, Michael L Nelson, and C Lee Giles. Modeling Updates of Scholarly Webpages Using Archived Data. In *Proceedings of IEEE International Conference on Big Data (IEEE BigData 2020)*, pages 1868–1877, 2020.

[10] Nwala, Alexander C, Michele C Weigle, and Michael L Nelson. Using Micro-collections in Social Media to Generate Seeds for Web Archive Collections. In *Proceedings of ACM/IEEE Joint Conference on Digital Libraries (JCDL 2019)*, pages 251–260, 2019. (**Best paper nominee**).

[11] Nwala, Alexander C, Michele C Weigle, and Michael L Nelson. Scraping SERPs for Archival Seeds: It Matters When You Start. In *JCDL '18: Proceedings of the 17th ACM/IEEE-CS Joint Conference on Digital Libraries*, pages 263–272, 2018.

[12] Nwala, Alexander C, Michele C Weigle, and Michael L Nelson. Bootstrapping Web Archive Collections from Social Media. In *HT '18: Proceedings of ACM ACM Conference on Hypertext and Social Media*, pages 64–72, 2018.

[13] Shawn M Jones, Nwala, Alexander C, Michele C Weigle, and Michael L Nelson. The many shapes of Archive-It. In *Proceedings of iPres*, 2018.

[14] Grant Atkins, Nwala, Alexander C, Michele C Weigle, and Michael L Nelson. Measuring News Similarity Across Ten U.S. News Sites. In *Proceedings of iPres*, 2018.

[15] Nwala, Alexander C, Michele C Weigle, Michael L Nelson, Adam B Ziegler, and Anastasia Aizman. Local Memory Project: Providing tools to build collections of stories for local events from local sources. In *JCDL '17: Proceedings of the 17th ACM/IEEE-CS Joint Conference on Digital Libraries*, pages 219–228, 2017.

[16] Nwala, Alexander and Michael L Nelson. A supervised learning algorithm for binary domain classification of web queries using SERPs. In *JCDL '16: Proceedings of the 16th ACM/IEEE-CS Joint Conference on Digital Libraries*, pages 237–238, 2016. (**Best Poster Award, 3rd place**).

Service

William & Mary

- **University (2024 - 2025):** Quality Enhancement Plan Committee (QEP - Infrastructure Working Group), Vice Provost for Enrollment Strategy search committee, and Camp Launch (Focusing on the Future) Mentor.
- **Data Science Committees:** 2024: Curriculum committee member. 2022 – 2023: Nominations and awards committee, Search committee for Assistant/Associate/Professor of Data Science and Director of the CGA.
- **Recommender:** 2022: Ethan Nelson (grad. school) and Sofia Huang (grad. school). 2023: Suditi Shyamsunder (Botetourt award) and Sampath Jayarathna (Tenure). 2024: Gangani Ariyarathne (NSF travel award)

- **Co-lab instructor (Fall 2023/2024):** Anthologies-Canons; ENGL 480-01 (Primary Instructor: Arthur Knight)
- **Collaborators:** 2023 – 2024: Arthur Knight - co-instructed Anthologies Lab and instructed W&M students to develop tool to automate data entry. Collaborated on Network Science course group project based on Dr. Knight's research interest. 2024 – present: Jaime Settle - collaborating on NSF HNDS-I grant. James Alcorn (Director of the Election Law Program at W&M Law School) - collaborating on developing machine-learning models to classify election law legal documents.

Extramural

- **Journal Editor (2024 – Present):** PeerJ, Computer Science
- **PC Member/Peer reviewer (2020 – Present):** ACM/IEEE Joint Conference on Digital Libraries (JCDL), ACM Web Science
- **NSF proposal reviewer (2023):** NSF CISE
- **PC Member/Peer reviewer (2023):** iPRES2023, EPJ Data Science, TheWebConf2023-Companion (CySoc 2023)
- **Tutorial/Workshop/Conference session Chair (2021/2023):** ACM/IEEE JCDL
- **Journal Reviewer (2020):** PLoS ONE
- **Collaborators:** 2023 – present: Fil Menczer and OSoMe (Indiana University) - collaborated to develop coordiscope and Information Operations dataset. 2024 – present: Michael Nelson and Michele Weigle (Old Dominion University) - collaborating on NSF HNDS-I grant. Faryaneh Poursardar (Old Dominion University) - collaborating on Minerva (DoD) grant.

Selected Invited Presentations & Press

2026-03-18	Measuring Digital Local News Desertification: Presented to PRSA Hampton Roads at Fort Monroe Hampton
2024-07-{20, 22, 24}	NSF service: Presented to high school students at W&M's Camp Launch's Focusing on the Future
2024-09-25	Presented agwagram to Observatory on Social Media at Indiana University
2023-10-11	Panelist/paper reviewer of GRI Research in Progress Seminar: "Networks of the Belt & Road: The Hidden Role of Financial Brokers"
2023-09-13	Press: William & Mary News - Modeling social media behaviors to combat misinformation
2023-06-17	Presented "A Language Framework for Characterizing Information Operations" to ODU summer REU students at the invitation of Dr. Jian Wu
2023-04-23	Presented "A Language Framework for Characterizing Information Operations" to CS800 at ODU at the invitation of Dr. Stephen Olariu
2023-04-19	Presented "Vectors Of Online Information Manipulation at Reading@Russia" at the invitation of Dana Willner
2022-10-26	Data Science representative at STEM & Social Science Undergraduate Research Fair
2022-09-21	Panelist on IGNITE talk for prospective minority faculties
2022-04-06	Press: PBS - Is the internet expanding or narrowing our minds?
2021-10-13	StoryGraph: quantifying the attention of news stories through near-real time story link detection , Knight Foundation Tool Demo
2021-09-28	Leveraging the domain expertise of social media users to generate seeds for web archive collections , CS Colloquium, Old Dominion University (ODU), Norfolk, VA
2021-08-28	Garbage, Glitter, or Gold: Assigning Multidimensional Quality Scores to Social Media Seeds for Web Archive Collections , ACM/IEEE JCDL 2021
2021-03-25	OSoMe Tools , Bloomington High School, Bloomington, IN
2021-02-10	365 Dots in 2018, 2019, and 2020: Quantifying Attention of News Sources , Computational + Journalism Symposium, Northeastern University, Boston, MA
2021-02-02	My Research Story/Experience , CS 800, ODU
2020-08-05	My Research Experience , Web Archiving And Digital Libraries Workshop Panel
2020-05-18	Bootstrapping Web Archive Collections From Micro-collections In Social Media , Virginia Modeling, Analysis & Simulation Center, Suffolk, VA
2020-03-26	Bootstrapping Web Archive Collections From Micro-collections In Social Media , University of Denver, Denver, CO

2019-08-21 Masters/PhD program survival guide , ODU

2019-06-04 Using Micro-collections in Social Media to Generate Seeds for Web Archive Collections , ACM/IEEE JCDL 2019, Urbana-Champaign, IL

2019-04-29 Bootstrapping Web Archive Collections From Micro-collections In Social Media , PhD Candidacy Proposal, ODU

2018-07-11 Bootstrapping Web Archive Collections from Social Media , ACM Hypertext and Social Media (HT 2018), Baltimore, MD

2018-06-05 Scraping SERPs for Archival Seeds: It Matters When You Start , ACM/IEEE JCDL 2018, Fort Worth, TX

2017-08-04 Primitive analyses for studying media manipulation , Harvard Library Innovation Lab, Cambridge MA

2017-07-27 Introduction to Programming with Python , Harvard University Skill Share, Cambridge MA

2017-06-21 Local Memory Project , ACM/IEEE JCDL 2017, Toronto ON, Canada

Publications - Selected Blog posts

2021-05-10 Chronicling the life-cycle of top new stories with StoryGraphBot

2020-01-20 366 dots in 2020 - top news stories of 2020

2020-09-28 My report card to my mother

2020-08-31 DNC vs RNC pulses - Quantifying news attention for the DNC and RNC with StoryGraph

2020-08-10 StoryGraph, reading the news for three years, a look at the past and the future

2020-01-04 365 dots in 2019 - top news stories of 2019

2019-09-09 Introducing sumgram, a tool for generating the most frequent conjoined ngrams

2019-06-05 Joint Conference on Digital Libraries (JCDL) 2019 Trip Report

2019-03-05 365 dots in 2018 - top news stories of 2018

2018-08-14 Revealing Data: Why We Need Humans To Curate Web Collections

2018-07-18 HyperText and Social Media (HT) Trip Report

2018-06-08 Joint Conference on Digital Libraries (JCDL) 2018 Trip Report

2018-05-04 An exploration of URL diversity measures

2018-03-04 Installing Stanford CoreNLP in a Docker Container

2017-08-27 Media Manipulation research at the Berkman Klein Center at Harvard University Trip Report

2017-06-29 Joint Conference on Digital Libraries (JCDL) 2017 Trip Report

2017-04-18 Local Memory Project - going global

2017-03-20 A survey of 5 boilerplate removal methods

2017-01-23 Finding URLs on Twitter - A simple recommendation

2016-11-16 Introducing the Local Memory Project

2016-10-26 A look back at the 2008 and 2012 US General Elections via Web Archives

2016-09-09 Summer Fellowship at the Harvard Library Innovation Lab Trip Report

2016-07-18 Tweet Visibility Dynamics in a Tweet Conversation Graph

2016-06-23 Joint Conference on Digital Libraries (JCDL) 2016 Trip Report

2016-05-31 Can I find this story? API: Yes, Google: Maybe, Native Search: No

2016-03-07 Archives Unleashed Web Archive Hackathon Trip Report (#hackarchives)

2015-07-22 I Can Haz Memento

2015-02-05 What Did It Look Like?

2014-11-14 Carbon Dating the Web, version 2.0